

Connection between Completeness and MVUE

- Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x; \theta)$.
- $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic with pdf $f_{Y_1}(y_1; \theta)$
- Y_2 is an unbiased estimator of θ such that $\mathbb{E} Y_2 = \theta$

- Define

$$\varphi(Y_1) = \mathbb{E}(Y_2|Y_1).$$

Then, $\mathbb{E}\varphi(Y_1) = \mathbb{E}Y_2 = \theta$

- If there is another unbiased estimator $\psi(Y_1)$ which is also a function Y_1 , then,

$$\mathbb{E}(\varphi(Y_1) - \psi(Y_1)) = 0 \quad \forall \theta$$

- If $\{f_{Y_1}(y_1; \theta) : \theta \in \Omega\}$ is a complete family, we have

$$\varphi(Y_1) = \psi(Y_1)$$

- For every other unbiased estimator $\psi(Y_1)$ of θ , we always have $\varphi(Y_1) = \psi(Y_1)$. That means that $\varphi(Y_1)$ is the MVUE of θ .

- Let $X_1, \dots, X_n \stackrel{iid}{\sim} f(x; \theta)$.
 - $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic
 - $\{f_{Y_1}(y_1; \theta) : \theta \in \Omega\}$ is a complete family

If $\varphi(Y_1)$ is an unbiased estimator θ , then $\varphi(Y_1)$ is the unique MVUE of θ .

Y_1 is a complete and sufficient statistic of θ .